

Explainable Bangla Document Classification Using Attention-Based Deep Learning Models

I. CONCLUSION, LIMITATIONS, AND FUTURE WORK

A. Conclusion

The paper proposes the explainable pipeline for Bangla newspaper document classification on 100K documents classified under 9 different categories with the same preprocessing, tokenization and training process using three neural networks Convolutional Neural Network (CNN), Bidirectional LSTM (BiLSTM) and Attention-BiLSTM (Attention BiLSTM). Overall, the Attention-BiLSTM performed best with 94.36% test accuracy, 0.1999 test loss, and 0.90 macro-averaged F1-score, although it was not the deepest network, but the same recurrent feature extractor was used in both of these models. Although these three models are not deepest, the plain BiLSTM turned out to be lighter than the CNN baseline, and the smallest test loss (0.1999), the highest accuracy (94.36%) and macro-averaged F1 score (0.90) compared to the CNN baseline. The advantage was also not evenly distributed, since, when studying the data per class, all three models showed very high accuracy on the majority classes of *bangladesh* and *sports* with the attention model showing the highest gains on the minority classes, such as lexically diffuse classes as *opinion* and *life-style* which were harder to label. Analysis of the training and validation curves also revealed that all three models start to overfit in the first 2-3 epochs, thus highlighting that the selection of the best epoch to check-point, and early stopping, and not the training time, were the key to get to the test performance reported. Finally, the saliency mechanism embedded in the Attention-BiLSTM model was used to calculate the gradient of the top words per category, with words that were intuitive and intuitively aligned with each category that was predicted for the token-level explanations being used as a useful instance-level auditing tool in a language where similar tools are few and far between. All of these results suggest that a well-designed attention enhanced recurrent model can be as accurate as or more accurate than the more complex model, on formal news text in Bangla, with a lower computational cost than a large pretrained transformer, and that a gradient-based explanation technique is a viable approach for developing formal text classification systems that are both accurate and audit worthy.

B. Limitations

It is important to recognize that the present study has a number of limitations. Second, the size of the data set is large in absolute size (100,000 articles) but is not balanced across categories, particularly the class *bangladesh* where

the class has a large majority, which may result in higher accuracy than macro-averaged values, and may underestimate the effort needed to deploy minority class categories like *economy* and *life-style* with different class distributions. Secondly, the models have been trained and evaluated on formally written newspaper text, and performance has not been tested when evaluated on informal, code-mixed and social media Bangla, which is very different from the text sources they are trained from, both in terms of vocabulary, spelling consistency and sentence structure, as some similar results have been observed on transformer-based Bangla classifiers when tested on out-of-domain text sources. Lastly, in this study, a fixed-size word vocabulary of 50wk is used for the word-level tokenizer, potentially limiting its efficiency in representing out-of-vocabulary word variants and neologisms when compared to subword and character-level tokenization methods. Fourth, the large pretrained transformer models such as BanglaBERT and XLM-RoBERTa, which have performed well in the other Bangla classification tasks, are not included in the comparison in this paper intentionally to assess the attention mechanism performance in a class of smaller models, but not necessarily the best results that can be achieved with transformer models. Lastly, the explainability analysis performed in this work is qualitative and instance-level, which is to say, no human-subject evaluation, or a quantitative faithfulness metric (such as deletion/insertion curves, comprehensiveness or sufficiency scores) was applied to ensure that the returned saliency scores are indeed the actual decision process of the model, and not an approximation of it that looks plausible to human readers.

C. Future Work

Based on these results and limitations, there are a few directions in which to proceed in future research. For one, by adding the pretrained transformer models like BanglaBERT, mBERT, and XLM-RoBERTa, the differences between the computational efficiency of the attention-augmented recurrent model proposed here and the potentially superior accuracy of the large pretrained language models especially in minority categories could be understood. Second, there are other word-level tokenizers available that can have a variable vocabulary, like subword tokenization (e.g., Byte-Pair Encoding or SentencePiece), which may be more robust to Bangla's rich morphology and less susceptible to out-of-vocabulary loss. Thirdly, the pipeline evaluation on non-formal corpora like informal text, code-mixed text and Social Media text would allow to check the generalizability of the pipeline models and the explainability mechanism beyond the formal written

news articles. Finally, further studies on how to measure the faithfulness of explanations in a quantitative manner, alongside the qualitative analyses presented here, and if possible, with a human evaluation of the saliency of the explanations, would allow to validate trustworthiness of the model's stated reasoning further. Finally, directly addressing class imbalance (by oversampling, class weighted loss functions, or focal loss), may further enhance results on the most difficult categories in this study, which are the underrepresented categories. Finally, the approach would be shown to be useful beyond off-line benchmark evaluation, by being deployed in a real world system for news categorisation or content moderation and continuously monitored in production with the explanations based on saliency.